ECE 417 / ECE 613: Image Processing and Visual Communication

Final Project Instructions

Winter, 2026

1. Project Overview

The final project consists of a single, integrated investigation in the area of **image processing and visual communication**, with two required components:

1. **Reproducible Component**
2. **Novelty Component**

The objective is to develop strong skills in understanding existing research, reproducing published results, and extending or critically analyzing those results through original experimentation.

Each project must include:

- A focused review of relevant literature
- Careful implementation and reproduction of existing methods
- Experimental or computational work
- A novel extension, investigation, or insight beyond simple reproduction
- Critical analysis and discussion

2. Team Formation and Project Registration

Projects may be completed individually or in teams of **up to three students**.

Each team must send a brief email to the instructor indicating:

- Names and student numbers of all team members
- Whether the project is individual or team-based
- A short description of the proposed project topic or application area

This information is required to register the project. Project topics may be refined as the term progresses.

3. Project Components

3.1. Reproducible Component

In this component, teams must select one or more relevant research papers and **reproduce key results** using their own implementation.

This includes:

- Clear description of the chosen method(s)
- Proper citation of the original work
- Implementation details and experimental setup
- Comparison between reproduced results and those reported in the paper
- Discussion of discrepancies, limitations, and practical challenges

The emphasis is on understanding and faithful reproduction, not on achieving identical numerical results.

3.2. Novelty Component

Building on the reproducible component, teams must include a **novel contribution**. This may take one of several forms, including but not limited to:

- Modifying or extending an existing algorithm
- Applying the method to a new dataset or problem
- Comparing multiple methods to gain new insights
- Performing ablation studies or robustness analyses
- Proposing and evaluating a new idea inspired by the literature

Novelty does not require inventing an entirely new algorithm. Well-motivated analysis and insightful experimentation are equally valued.

4. Project Scope and Topic Selection

Projects must fall within the broad scope of image processing and visual communication. Example areas include:

- Image enhancement, restoration, and denoising
- Image and video compression
- Medical and biomedical imaging
- Computer vision and visual recognition
- Visual communication systems
- Learning-based image and video processing

Projects should be well scoped and technically focused. Overly broad surveys or purely incremental parameter tuning are discouraged.

5. Use of Learning-Based Methods

Projects involving deep neural networks or other learning-based methods are permitted. However:

- Simply retraining or fine-tuning existing models without analysis is insufficient.
- Greater emphasis is placed on motivation, reasoning, experimental design, and interpretation rather than raw performance.
- Thoughtful use of pre-trained models, controlled experiments, and comparative analysis is encouraged.

6. Final Report Format

Each team must submit a **single final report** that integrates both the reproducible and novelty components.

Formatting requirements:

- IEEE Signal Processing Society conference style (ICIP-style)
- Maximum length: **6 pages**, including figures and tables
- References may appear on an additional page if needed

The report should include:

- Title, authors, and affiliations
- Abstract and keywords
- Introduction and background
- Reproducible component (methods and results)
- Novelty component (extensions and analysis)
- Discussion and conclusions
- References

Students are encouraged to consult recent ICIP papers for structure and style.

7. Submission Details

- **Final report due: April 24, 2026**
- **Submission method: PDF upload via Learn**
- Only one submission per team is required
- No in-class presentations will be held

Late submission policies follow the course outline.

8. Academic Integrity

All submitted work must be original and properly cited.

- Copying text, figures, or tables without attribution is not permitted.

- Plagiarism detection software may be used.
- Relevant university academic integrity policies apply.